

# Latent Residual Diffusion for High-Quality LiDAR Point Cloud Completion

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**Abstract**—Autonomous driving technology relies on high-quality LiDAR data, yet the high cost of LiDAR sensors limits its widespread adoption. In large-scale outdoor scenes, the inherent sparsity and irregularity of point clouds result in significant structural information loss, further complicating the efficient use of LiDAR data. To address this, we propose an innovative Residual Diffusion Model for high-quality point cloud completion. Unlike existing methods that struggle with the trade-off between structure and detail, our model operates in a latent space to decouple the completion task. By jointly diffusing structural residuals with random noise, it enables the collaborative reconstruction of both macroscopic geometry and fine-grained details during the reverse process. This process is optimized by a comprehensive loss function that integrates residual prediction, noise distribution regularization, and latent space smoothness constraints. Our method exhibits superior generative capabilities in experiments on the Semantic-KITTI and VOD dataset, achieving excellent results in shape detail completion.

**Index Terms**—LiDAR completion, point clouds, diffusion model, 3D sparse convolution.

## I. INTRODUCTION

Environmental perception is fundamental to autonomous driving, enabling vehicles to sense and interpret surrounding scenes through integrated sensor systems. Among these, LiDAR is indispensable for constructing high-resolution 3D representations by measuring laser signal travel time, thus providing accurate geometric and depth information crucial for navigation and decision-making in complex environments [1]–[4]. However, LiDAR’s high cost and scalability limitations hinder widespread adoption. Consequently, techniques enhancing the utility of existing LiDAR data are in high demand. A key research area is LiDAR point cloud completion, which reconstructs missing or incomplete parts to improve point cloud density and structural integrity.

The task of completing and enhancing LiDAR point clouds is multifaceted. Traditionally, efforts have focused on upsampling to increase point density and data augmentation to create more diverse training sets. Early upsampling methods such as PU-Net [5] utilized multi-branch convolutional learning to generate uniformly distributed points, while later works

adopted generative models. PU-GAN [6] employs a Generative Adversarial Network (GAN) for training, comprising a generator and discriminator. PUFA-GAN [7] refined this by incorporating frequency-aware discrimination for more realistic high-frequency details. Nonetheless, GAN training is inherently unstable and susceptible to mode collapse, often resulting in reduced diversity and geometric inaccuracies.

Another significant research direction involves realistic scene generation, which aims to create entirely new, plausible LiDAR scenes. This is often addressed by combining RGB images with LiDAR point clouds. In this approach, RGB images are used to infer depth maps, while LiDAR data provides supervision [8], [9]. However, this can be limited by the quality of RGB images and potential perspective mismatches, leading to inaccuracies. Other methods use signed distance fields (SDF) [1], [10], [11] to represent scenes as voxel grids, with each voxel capturing the distance to the nearest surface. While this method approximates surfaces, it often loses finer details due to the limited resolution of voxels.

More recently, Diffusion Models (DMs) have recently shown remarkable capability in high-quality image synthesis [12]–[14] by progressively denoising data. Motivated by this success, researchers have applied DMs to LiDAR point cloud generation and completion in real-world scenarios [15]–[19], often by converting point clouds into 2D range images to exploit mature 2D diffusion frameworks. LiDMs [17] employ a curve-based generator with latent space compression for conditional LiDAR scene synthesis; LiDARGen [20] trains range and reflectance modalities jointly using score-based models such as NCSNv2 [21]; R2DM [15] generates high-quality range and reflectance data; and RangeLDM [16] integrates Hough Voting within a latent diffusion pipeline to model range-view distributions for fast scene generation.

However, these diffusion-based methods relying on range images often lose essential 3D geometric information, limiting fine-grained reconstruction accuracy. To mitigate this, we propose a Residual Diffusion Model that operates in 3D latent space, where a residual formulation captures the gap between partial and complete scenes for geometry-aware LiDAR completion. In summary, the key contributions of this paper are as follows.

- We propose a Residual Diffusion Model for point cloud

completion, introducing a joint residual-noise learning mechanism to effectively decouple structural recovery from fine-grained detail generation.

- Our framework utilizes 3D sparse convolution to operate directly on raw 3D data, preserving topological integrity and avoiding geometric information loss associated with 2D range-view projections.
- We introduce latent smoothness and noise regularization to ensure a continuous latent representation, enhancing the stability and fidelity of point cloud completion.

This paper is organized as follows. The proposed framework is introduced in Section II. We introduce our framework, including the sparse encoder, the residual diffusion module, and the refinement decoder for generating the completed point cloud. Experimental results achieved on the Semantic-KITTI and VOD dataset are presented in Section III.

## II. PROPOSED METHODOLOGY

### A. Architecture

Given a sparse partial point cloud  $\mathbf{P}$  representing a laser scan, the goal of point cloud completion is to predict a dense, complete geometry  $\mathbf{G}$  that recovers the underlying scene structure. As illustrated in Fig. 1, the pipeline consists of Sparse Encoder  $\mathcal{E}$ , Residual Diffusion Module, and a Refinement Decoder  $\mathcal{D}$ .

First, the encoder  $\mathcal{E}$  projects the input  $\mathbf{P}$  and the ground truth  $\mathbf{G}$  into a shared high-dimensional sparse latent space, yielding feature representations. To bridge the gap between partial and complete views, we reformulate the generation process as a dual-diffusion mechanism. Instead of diffusing the target features towards a zero-mean Gaussian distribution, we explicitly model the latent residual  $\mathcal{R} = \mathcal{Z}_G - \mathcal{Z}_P$ . In the forward process, structural residuals and random noise are jointly injected. Consequently, the reverse process is decoupled into two parallel tasks: predicting the deterministic residual  $\hat{\mathcal{R}}$  to recover macro-structures, and estimating the stochastic noise  $\epsilon$  to synthesize high-frequency details. Finally, the reconstructed features are decoded by  $\mathcal{D}$  to generate the final dense point cloud  $\hat{\mathbf{G}}$ .

### B. Diffusion-based LiDAR Completion

DMs are generative frameworks that synthesize data by gradually transforming Gaussian noise into structured outputs through a learned successive denoising process. Given a clean sample  $\mathbf{x}_0$ , the forward step at time  $t$  can be expressed as:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}), \quad (1)$$

where  $\bar{\alpha}_t$  is determined by the noise schedule  $\beta_t$ . In the reverse process, the noise is estimated by a structured convolutional neural network, typically adopting encoder-decoder designs such as U-Net, to progressively recover  $\mathbf{x}_0$  from noisy inputs.

However, unlike dense and uniformly distributed image or video data, LiDAR point clouds are intrinsically sparse and spatially irregular, making them incompatible with standard diffusion models that assume dense input structures. Traditional voxelization can regularize the points but inevitably

introduces large computational overhead and geometry loss. To address these issues, we design a diffusion framework that works directly on raw point coordinates, preserving detailed geometry while avoiding the cost of voxelization. Formally, the process begins with the input sparse scan, denoted as  $\mathbf{P} \in \mathbb{R}^{N \times 3}$ . To establish a supervision target that captures the complete scene geometry, the ground truth  $\mathbf{G} \in \mathbb{R}^{M \times 3}$  is constructed through temporal accumulation. Specifically, laser scans from a local temporal window—encompassing the preceding and subsequent 10 frames aligned to the current ego-pose—are superimposed. This aggregation creates a semi-dense structural reference that fills in occlusions naturally present in the single-frame view.

Both the partial input  $\mathbf{P}$  and the accumulated scene  $\mathbf{G}$  are mapped into a shared feature space using a sparse encoder  $\mathcal{E}$ . Structurally, the encoder is composed of six stacked sparse ResNet blocks, with each block incorporating strided sparse convolutions for downsampling. These sparse convolutions operate exclusively on the effective points within the input, thereby enhancing computational efficiency while maintaining significant geometric information. Each sparse ResNet block consists of a sparse convolution layer, followed by a nonlinear activation ReLU function to introduce nonlinearity, and a normalization layer to stabilize the learning process. Consequently, the encoder yields the latent representations:  $\mathcal{Z}_P = \mathcal{E}(\mathbf{P})$ ,  $\mathcal{Z}_G = \mathcal{E}(\mathbf{G})$ . These representations are formulated as sparse Tensors, consisting of a set of coordinate indices and associated feature vectors.

With the latent representations established, we formulate the diffusion generative process within this feature space. To mitigate the inefficiency of synthesizing geometry from pure Gaussian noise and to fully leverage the structural priors in the partial observation  $\mathcal{Z}_P$ , we introduce a latent residual term, denoted as  $\mathcal{R} = \mathcal{Z}_G - \mathcal{Z}_P$ . This term explicitly characterizes the feature disparity between the complete scene  $\mathcal{Z}_G$  and the partial reference  $\mathcal{Z}_P$ . Building on this, we reformulate the forward diffusion process. Specifically, we model the forward transition as a joint process of deterministic feature erosion and stochastic noise injection. At any arbitrary time step  $t$ , the latent state  $\mathcal{Z}_t$  is derived directly from the complete representation  $\mathcal{Z}_G$  via the following closed-form formulation:

$$\mathcal{Z}_t = \mathcal{Z}_G - (1 - \sqrt{\bar{\alpha}_t}) \mathcal{R} + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}). \quad (2)$$

Here, the term  $-(1 - \sqrt{\bar{\alpha}_t}) \mathcal{R}$  acts as a deterministic drift term, while  $\sqrt{1 - \bar{\alpha}_t} \epsilon$  represents the stochastic noise injection. This formulation provides a mathematically rigorous interpolation between the two states:

- At  $t = 0$ , since  $\bar{\alpha}_0 = 1$ , the drift and noise terms vanish, yielding  $\mathcal{Z}_0 = \mathcal{Z}_G$ .
- As  $t \rightarrow T$ ,  $\bar{\alpha}_T \approx 0$ . The equation becomes  $\mathcal{Z}_T \approx \mathcal{Z}_G - \mathcal{R} + \epsilon$ . Given our definition  $\mathcal{R} = \mathcal{Z}_G - \mathcal{Z}_P$ , this simplifies to  $\mathcal{Z}_T \approx \mathcal{Z}_P + \epsilon$ .

In the reverse diffusion process, our goal is to convert the noisy latent  $\mathcal{Z}_T$  into a complete scene  $\mathcal{Z}_G$ . We employ a dual-head design based on Minkowski Networks, focusing on two tasks:

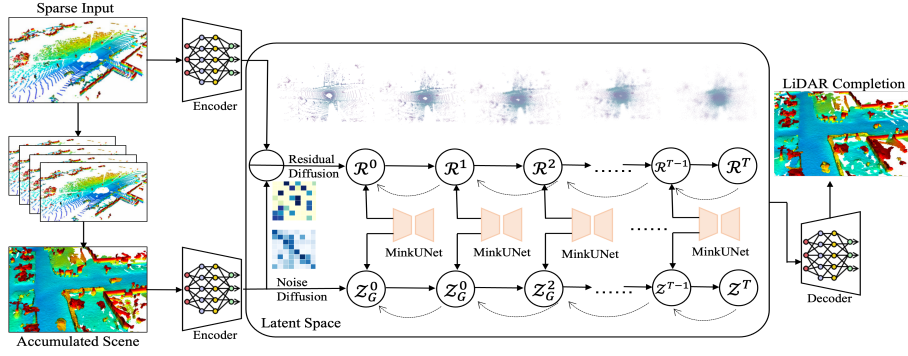


Fig. 1: Overview of proposed 3D point cloud completion architecture.

predicting residual  $\hat{\mathcal{R}}$  and estimating noise  $\epsilon$ . The first task uses sparse convolutions to recover  $\hat{\mathcal{R}}$ , capturing the scene's macro-structures. The second task estimates  $\epsilon$  to synthesize details, ensuring realism and detail richness in the point cloud. The network architecture features sparse convolutional layers, integrated with downsampling, upsampling modules, and skip connections to retain detail information.

The decoder efficiently transforms the latent feature representation  $\mathcal{Z}_0$  into a complete point cloud, achieving effective spatial information recovery. Utilizing four layers of sparse deconvolution, the decoder progressively enhances spatial resolution, ensuring the effective capture of multi-scale features. Through this approach, the decoder's final output, reconstructed point cloud  $\hat{\mathbf{G}}$ , accurately reflects the geometric structure of the scene, excelling in detail retention and overall consistency.

In the inference phase, the sparse reference frame  $\mathbf{P}$  containing  $N$  points is used as input. These points are replicated  $k$  times to match the number  $M$  of the ground truth  $\mathbf{G}$ , ensuring dimensional compatibility. This input is encoded into a latent representation using a sparse encoder. A reverse diffusion process iteratively refines the latent representation derived from the partial input, gradually transforming it into a complete and coherent scene description. The decoder then translates these refined features into the final point cloud  $\hat{\mathbf{G}}$ , ensuring the final output accurately reflects the geometric structure of the original scene.

### C. Loss Function

In this section, we define a multi-objective loss function designed to optimize point cloud completion. Our approach balances noise estimation, latent feature mapping, and geometric stability.

During the reverse diffusion process, the network must accurately estimate the Gaussian noise injected into the samples during the forward pass. This is captured by the diffusion loss

$$\mathcal{L}_{\text{diff}} = \|\epsilon - \hat{\epsilon}_\theta\|_2^2, \quad (3)$$

where  $\epsilon$  is the random noise injected during the forward diffusion process, and  $\hat{\epsilon}_\theta$  represents the noise estimated by the model at time step  $t$ .

To bridge the gap between partial inputs and complete scenes, we introduce a residual prediction loss. This term focuses on the latent feature differences between the observed geometry and the target completion

$$\mathcal{L}_{\text{res}} = \|\mathcal{R} - \hat{\mathcal{R}}\|_2^2, \quad (4)$$

where  $\mathcal{R}$  is the latent representations of the complete scene and partial input, respectively, and  $\hat{\mathcal{R}}$  is the model's predicted residual.

The smoothness loss is designed to enforce continuity in the latent space, preventing geometric artifacts or discontinuities in the reconstructed point cloud:

$$\mathcal{L}_s = \mathbb{E}_\epsilon \|\nabla_\epsilon (\sqrt{1 - \alpha_t} \cdot \hat{\epsilon}_\theta \cdot \Delta \hat{\epsilon}_\theta) - a \cdot \Delta \epsilon\|_2^2. \quad (5)$$

This loss constrains the impact of noise variation on signal generation, ensuring smoothness and stability in the generated point cloud.

The total loss is a weighted sum of the aforementioned terms, governed by the hyperparameters  $r_1$  and  $r_2$ :

$$\mathcal{L} = \mathcal{L}_{\text{diff}} + r_1 \mathcal{L}_{\text{res}} + r_2 \mathcal{L}_s. \quad (6)$$

where  $r_1$  and  $r_2$  are hyperparameters that adjust the weights of different loss terms.

## III. EXPERIMENTAL RESULTS

### A. Datasets and Implementation Details

In this section, two datasets are used for performance evaluation.

- Semantic-KITTI Dataset [22]: Part of the KITTI Vision Benchmark Suite, provides over 43,000 densely annotated LiDAR scans covering diverse urban and rural driving scenes. It includes 25 semantic classes, enabling comprehensive analysis of complex driving environments.
- VOD Dataset [23]: Provides 8,600 frames of synchronized and calibrated data from 64-layer LiDAR, stereo cameras, and 3 + 1D radar, collected in complex urban traffic scenes in Delft, Netherlands. It is designed to support research in autonomous driving and robotic perception.

TABLE I: Quantitative Comparisons on the two Datasets.

| Method        | Semantic-KITTI |             |             | VOD         |             |             |
|---------------|----------------|-------------|-------------|-------------|-------------|-------------|
|               | CD↓            | HD↓         | F-Score↑    | CD↓         | HD↓         | F-Score↑    |
| LiDARGAN [24] | 4.72           | 10.44       | 0.62        | 5.89        | 14.02       | 0.58        |
| LiDARVAE [25] | 2.71           | 8.27        | 0.67        | 4.35        | 14.29       | 0.58        |
| LiDARGen [20] | 1.55           | 9.22        | 0.71        | 3.79        | 9.50        | 0.63        |
| RangeLDM [16] | 1.25           | 8.42        | 0.76        | 3.65        | 8.66        | 0.64        |
| LiDM [17]     | 0.96           | 6.75        | 0.73        | 2.85        | <b>7.28</b> | 0.70        |
| LiDiff [18]   | <b>0.53</b>    | 7.30        | 0.78        | 2.60        | 7.80        | 0.74        |
| None residual | 0.70           | 6.59        | 0.81        | 2.59        | 7.98        | 0.75        |
| Ours          | 0.67           | <b>6.10</b> | <b>0.85</b> | <b>2.26</b> | 7.71        | <b>0.78</b> |

Our model is trained for 100 epochs with a batch size of 64. Both models are optimized using the Adam optimizer with an initial learning rate  $9 \times 10^{-4}$ . Training and evaluation are performed using PyTorch on Nvidia GTX 3090 GPU.

### B. Performance Comparison

To quantitatively evaluate the geometric fidelity of the generated point clouds against the ground truth, we employ three standard metrics: Chamfer Distance (CD), Hausdorff Distance (HD), and F-Score.

- CD assesses the overall shape similarity by calculating the average nearest-neighbor distance between points in the generated and ground truth sets.
- HD measures the maximum nearest-neighbor distance between two point sets, capturing worst-case deviations and highlighting outliers.
- F-Score provides a comprehensive measure of reconstruction quality by computing the harmonic mean of precision and recall at a specific distance threshold.

Table I compares our method with state-of-the-art baselines on the Semantic-KITTI and VOD datasets. Compared to the strongest competitor, LiDiff, our approach demonstrates superior structural fidelity and robustness. On Semantic-KITTI, while achieving a comparable CD, our method significantly reduces the outlier-sensitive HD by 1.20 and improves the F-Score by 0.07, indicating much cleaner geometric boundaries. This advantage is further amplified on the VOD dataset, where our model outperforms LiDiff with a clear margin, reducing CD by 0.34 and increasing F-Score by 0.04. These results confirm that our method generates point clouds with higher precision and completeness across different environments. The ablation results in the last two rows further validate the effectiveness of our proposed design. By integrating the residual diffusion module, the model achieves consistent performance gains over the baseline without residual learning. Specifically, the inclusion of this module drives a 0.33 reduction in CD on the VOD and a 0.04 uplift in F-Score on Semantic-KITTI.

To provide a more intuitive assessment of the generation quality, we conduct visualization experiments on the VOD and Semantic-KITTI datasets. Fig. 2 illustrates the completion results on the VOD dataset. As observed, our method reconstructs globally consistent geometric layouts from sparse inputs, accurately restoring road surfaces, building facades, and surrounding vegetation without introducing scattered artifacts. Fig. 3 further provides a detailed comparison on the

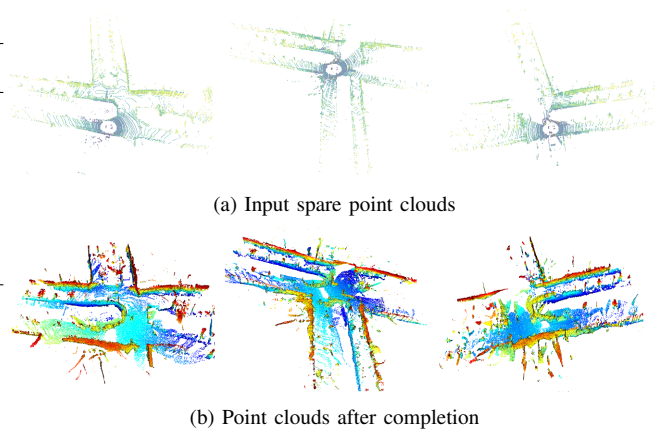


Fig. 2: LiDAR completion on VOD dataset.

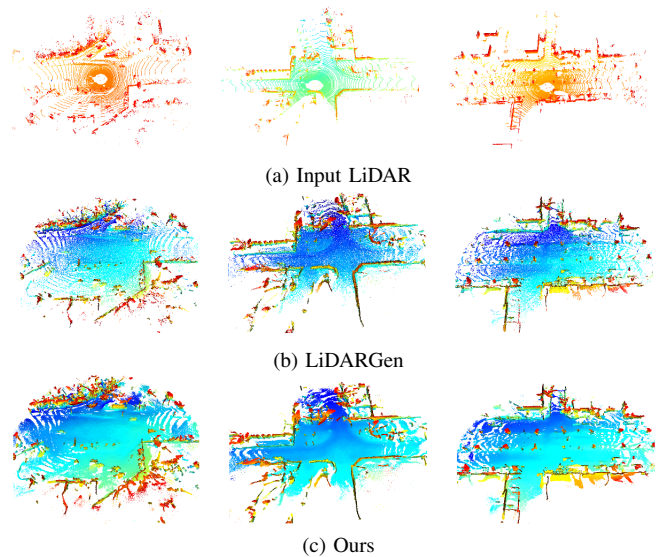


Fig. 3: LiDAR completion on Semantic-KITTI dataset.

Semantic-KITTI dataset. Fig. 3a shows the input sparse point cloud, while Figs. 3b and 3c correspond to the completion results from LiDARGen [20] and our model, respectively. LiDARGen tends to generate noisy and spatially inconsistent points, leading to blurred object contours and hollow regions. In contrast, our method generates significantly denser and more coherent point clouds.

## IV. CONCLUSION

We propose a residual-guided dual-diffusion framework for LiDAR point cloud completion that works directly on raw coordinates to preserve fine geometry. By modeling latent residuals and decoupling structural recovery from noise prediction, our method accurately reconstructs both global shapes and detailed features. Experiments on Semantic-KITTI and VOD show clear gains over state-of-the-art methods in accuracy and visual quality, producing dense and coherent point clouds across diverse scenes.

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